The innate defense mechanisms are non-specific in the sense that they are effective against a wide range of potentially infectious agents. It is otherwise known as **non-specific immunity** or **natural immunity.**

A number of innate defense mechanisms are operative non-specifically against a large number of microorganisms as shown in the **Table 8.1** and **Fig. 8.2.**

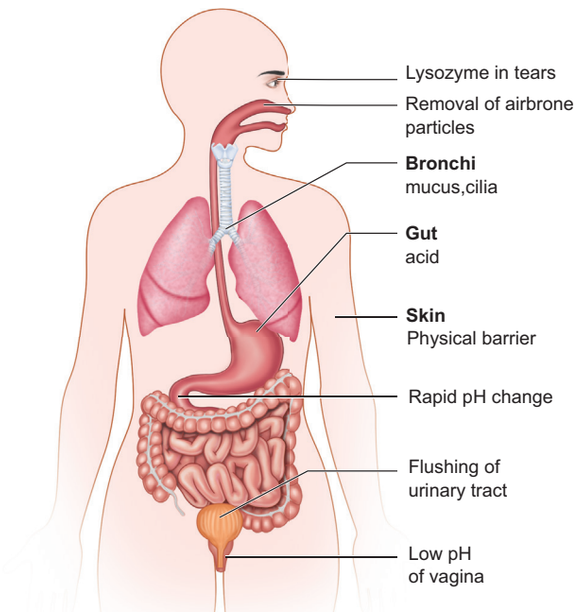


Fig. 8.2 Various anatomical and physiological barriers to microbial attack

8.3 Acquired immunity

The immunity that an individual acquires after birth is known as **acquired immunity**. It is the body's resistance to a specific pathogen.

The unique features of acquired immunity are antigenic specificity, diversity, recognition of self and non-self and immunological memory.

Components of acquired immunity

Acquired immunity has two components – **cell mediated immunity (CMI) and antibody mediated immunity or humoral immunity.**

1. Cell mediated immunity

When pathogens are destroyed by cells without producing antibodies, then it is known as cell mediated immune response or cell mediated immunity. This is brought about by T cells, macrophages and natural killer cells.

2. Antibody mediated immunity or humoral immunity

When pathogens are destroyed by the production of antibodies, then it is known as antibody mediated or humoral immunity. This is brought about by B cells with the help of antigen presenting cells and T helper cells. Antibody production is the characteristic feature of **vertebrates** only.

Types of acquired immunity

Acquired immunity may be **active immunity** or **passive immunity (Table 8.2)**.

The immunological resistance developed by the organisms through the production of antibodies in their body is called active immunity. Active immunity is acquired through the use of a person's immune responses, which lead to the development of memory cells. Active immunity results from an infection or an immunization.

Passive immunity does not require the body to produce antibodies to antigens. The antibodies are introduced from outside into the organism. Thus, passive immunity is acquired without the activation of a person's immune response, and therefore there is no memory.



The process of production of blood cells in the bone marrow is called **haematopoiesis**.

8.4 Immune responses

The immune responses may be **primary** or **secondary (Table 8.3)**.

Primary immune response

The primary immune response occurs when a pathogen comes in contact with the

XII Std Zoology Chapter 8 EM.indd 131 23/12/2022 16:43:14